

MATH 2211, SPRING 2026: HOMEWORK 7

- (1) Suppose that $A \in M_{m \times n}(F)$. For each of the following, prove the assertion or find a counterexample:
- (a) $\text{rank}(AB) \leq \text{rank}(A)$ for any $n \times p$ -matrix B (p arbitrary);
 - (b) $\text{rank}(BA) \leq \text{rank}(A)$ for any $p \times m$ -matrix B (p arbitrary).

Hint: It might be useful to convert to questions about images of linear transformations.

- (2) If $A \in M_{n \times n}(F)$ is an invertible matrix with inverse A^{-1} show that we have

$$AA^{-1} = I_n = A^{-1}A.$$

- (3) Show that the associated linear transformation

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$\begin{bmatrix} a \\ b \end{bmatrix} \mapsto \begin{bmatrix} 2a + b \\ a + 2b \end{bmatrix}$$

is diagonalizable.

- (4) Consider the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

Use the previous problem to compute A^n for any integer n .

Definition 1. The **transpose** of a matrix $A \in M_{m \times n}(F)$ with columns

$$A = [v_1 \quad v_2 \quad \cdots \quad v_n]$$

is the matrix $A^T \in M_{n \times m}(F)$ with rows given by

$$A^T = \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{bmatrix}$$

- (5) Show that the linear transformation $U : F^n \rightarrow F^m$ associated with A is surjective if and only if the linear transformation $U' : F^m \rightarrow F^n$ associated with A^T is *injective*.